

MASSKAR, Finland

WMO: 029010

Lat: 63.728N

Lon: 22.584E

Elev: 43

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-7.7	-18.1	2.0	-13.3	-11.6	3.0	-7.2	41.6	33.6	36.6	34.4	2.5	80	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	71.4	63.1	68.4	61.3	65.8	59.7	64.1	69.6	62.2	66.7	60.7	64.7	9.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	83.2	67.1	60.2	77.9	64.5	58.7	73.8	63.2	29.4	69.8	28.0	66.8	26.9	64.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.1	29.0	25.5	DB	-14.0	78.8	8.1	4.2	-19.8	81.8	-24.6	84.3	-29.1	86.7	-35.0	89.8	(4)
			WB	-14.3	67.4	8.0	3.2	-20.1	69.6	-24.8	71.5	-29.3	73.3	-35.1	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	16.4	18.3	26.0	34.5	44.7	53.7	59.8	57.4	49.7	42.0	31.9	23.9
	DBStd	16.65	14.01	12.59	8.56	6.64	6.19	5.69	4.19	4.99	4.74	5.29	8.09	9.62
	HDD50	5020	1041	888	743	467	188	25	0	4	61	250	544	809
	HDD65	9756	1506	1308	1208	915	628	343	171	239	458	712	994	1274
	CDD50	756	0	0	0	2	25	135	303	235	53	3	0	0
	CDD65	16	0	0	0	0	0	2	9	5	0	0	0	0
	CDH74	43	0	0	0	0	1	4	33	5	0	0	0	0
	CDH80	5	0	0	0	0	0	0	5	0	0	0	0	0
Wind	WSAvg	12.3	12.2	12.5	11.9	11.5	11.0	9.9	10.2	10.8	12.4	15.6	15.5	14.1
Precipitation	PrecAvg	20.8	1.5	1.1	1.1	1.0	1.5	2.0	2.6	2.2	2.0	2.2	1.9	1.7
	PrecMax	26.5	2.7	2.0	1.9	2.6	2.8	3.4	4.8	5.2	4.3	3.7	4.4	3.3
	PrecMin	15.2	0.4	0.1	0.2	0.2	0.2	0.5	0.2	0.8	0.8	0.6	0.2	0.7
	PrecStd	3.1	0.6	0.6	0.5	0.6	0.7	0.8	1.2	1.3	1.1	0.9	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	39.0	40.9	59.3	69.2	73.2	78.7	74.3	64.1	55.2	46.0	40.5
		MCWB	35.5	37.0	37.9	51.3	57.6	61.8	70.0	64.7	57.8	51.3	44.0	37.6
	2%	DB	35.9	37.4	38.0	51.6	62.4	68.6	71.9	70.4	60.3	52.6	44.1	38.2
		MCWB	33.7	35.7	35.2	44.0	54.1	59.9	64.0	63.3	56.3	49.9	42.4	36.5
	5%	DB	34.3	35.6	36.5	46.9	57.1	65.3	68.7	67.1	58.1	50.6	42.7	36.8
		MCWB	32.8	34.1	34.3	41.7	51.0	58.1	61.4	61.7	54.8	48.3	40.9	35.4
	10%	DB	32.9	33.8	35.3	43.5	53.3	62.1	66.2	64.7	56.4	49.0	41.3	35.4
		MCWB	31.8	32.5	33.4	39.8	48.2	55.8	60.1	60.0	53.2	46.9	39.5	34.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	37.6	38.2	51.7	58.6	63.7	71.0	66.6	59.3	53.3	44.6	38.4
		MCDB	37.1	38.5	39.7	58.7	67.3	70.3	77.3	70.9	62.7	54.8	45.5	39.3
	2%	WB	34.3	35.9	35.9	44.9	54.5	61.2	64.8	64.2	57.2	50.3	42.7	36.8
		MCDB	35.4	37.0	37.1	50.4	61.4	67.6	70.4	69.4	59.9	51.9	44.0	37.8
	5%	WB	33.0	34.1	34.7	42.3	51.6	58.7	62.4	62.2	55.6	48.6	41.2	35.7
		MCDB	34.0	35.2	36.0	46.3	56.6	64.2	66.7	66.5	57.5	50.2	42.4	36.6
	10%	WB	31.9	32.6	33.7	40.1	48.5	56.3	61.0	60.6	53.6	47.1	39.8	34.4
		MCDB	33.0	33.8	35.1	43.2	53.0	61.3	65.0	63.9	56.0	48.8	41.1	35.4
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.1	9.4	10.0	10.0	9.5	8.8	8.2	7.1	5.9	6.3	7.9
		MCDBR	6.5	5.2	8.0	14.9	16.1	13.6	12.9	12.3	9.4	6.7	5.3	5.8
		MCWBR	6.2	4.7	6.2	9.7	9.4	7.8	6.7	6.7	6.5	5.2	5.4	5.7
	5% WB	MCDBR	7.4	4.8	6.4	13.9	15.0	12.8	10.8	11.4	8.8	6.4	5.3	5.8
		MCWBR	7.2	4.6	5.4	9.4	9.4	8.3	6.6	6.5	6.6	5.3	5.5	5.8
Clear-Sky Solar Irradiance	taub	0.228	0.287	0.313	0.351	0.350	0.351	0.371	0.359	0.334	0.315	0.364	0.351	
	taud	2.022	2.151	2.213	2.272	2.440	2.480	2.449	2.478	2.526	2.373	2.617	2.571	
	Ebn at Noon	114	198	245	261	274	276	266	259	243	192	105	57	
	Edh at Noon	9	20	28	33	31	30	30	27	21	16	8	5	
All-Sky Solar Radiation	RadAvg	44	224	701	1205	1626	1795	1642	1253	718	268	62	18	
	RadStd	4	44	74	108	166	154	153	121	74	49	10	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page